

HIT 2.0: 草本成分靶点的增强平台

德宇燕^{1, †}, 根辉郑^{1, †}, 蔡翠王¹, 陈子昆¹, 天天毛¹, 建高^{2,3}, 余言¹, 湘义陈¹, 雪杰纪¹, 金玉余¹, 赛凤莫¹, 浩南温¹, 文昊汉¹, 孟迪周¹, 袁王¹, 王君¹, 唐凯林¹, * 及曹志伟⁴, *

¹中国上海同济大学生命科学与技术学院, 上海第十人民医院胃肠病学科, 邮编200092, ²国际人类表型研究院 (上海), 中国上海, ³复旦大学上海癌症中心胸外科, 上海, 中国, 以及 ⁴复旦大学生命科学学院, 上海200092, 中国

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摘要

文献中描述的草药成分靶点已被探索, 以促进草药的机制研究以及新药的研发。虽然有几个数据库提供了类似的资料, 但大多数仅限于2010年前的直面资料, 亟需更新。HIT 2.0 是作为最新策划数据集构建的, 聚焦于2000–2020年PubMed文献中的草本成分靶点。目前, HIT 2.0拥有10,031对复合-靶点活性对, 质量指标介于2208个靶点至1237种成分之间, 来自1250多种信誉良好的草药。分子靶点涵盖那些被直接/间接激活/抑制的/蛋白、蛋白结合剂以及酶底物或产物。还包括那些在单一成分处理下受调控的基因。交叉链接至TTD、DrugBank、KEGG、PDB、UniProt、Pfam、NCBI、TCM-ID等数据库。更重要的是, HIT可以实现自动T。

介绍

作为药物候选物的重要来源, 草本活性成分在新药开发中起着关键作用。从1981年到2019年, 据报道FDA批准的药物中有33.6%来源于天然产物或其衍生物 (1)。为了更好地理解草本化合物与分子靶点之间的相互作用, 首个草本成分-靶点数据库HIT于2010年建立, 通过人工整理3250篇文献中描述的1301个草本化合物靶点, 并提供便捷链接至治疗靶点数据库 (TTD) 和Drugbank等 (2,3)。HIT的靶点信息已被广泛用于研究天然化合物的机制, 并用于草药的发现。例如, 基于HIT, Luo Y等揭示了丹参酮在治疗肝癌中的治疗机制 (4)。同时, Wang等根据中医临床疗效识别了哮喘的潜在靶点。

在过去十年 (2011–2021) 中, 有关天然成分及其靶点的研究呈爆炸式增长。随后, 建立了多种涵盖草药化合物与靶点相互作用的数据库。值得注意的是, 一个优秀的天然产物示例数据库 NPASS (6) 提供了天然产物的实验测定的定量活性记录, 其中包括约 700 种草药成分的近 2,000 对草药成分-靶点配对。其他草药数据库也包含了草药成分-靶点配对的重要信息 (7–11), 但大多数数据库下载并整合了早期的 HIT 数据, 并附加了预测靶点, 如 HERB、TCMID、TCMSP 和 SymMap (7–10)。这些数据库极大地丰富了草药成分的靶点多样性。然而, 文献中描述的草药成分靶点

HIT 2.0: an enhanced platform for Herbal Ingredients' Targets

Deyu Yan^{①, †}, Genhui Zheng^{①, †}, Caicui Wang¹, Zikun Chen¹, Tiantian Mao¹, Jian Gao^{2,3}, Yu Yan¹, Xiangyi Chen¹, Xuejie Ji¹, Jinyu Yu¹, Saifeng Mo¹, Haonan Wen¹, Wenhao Han¹, Mengdi Zhou¹, Yuan Wang¹, Jun Wang¹, Kailin Tang^{①, *} and Zhiwei Cao^{4, *}

¹Dept. of Gastroenterology, Shanghai Tenth People's Hospital, School of Life Sciences and Technology, Tongji University, Shanghai 200092, China, ²International Human Phenome Institutes (Shanghai), Shanghai, China, ³Department of Thoracic Surgery, Fudan University Shanghai Cancer Center, Shanghai, China and ⁴School of Life Sciences, Fudan University, Shanghai 200092, China

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ABSTRACT

Literature-described targets of herbal ingredients have been explored to facilitate the mechanistic study of herbs, as well as the new drug discovery. Though several databases provided similar information, the majority of them are limited to literatures before 2010 and need to be updated urgently. HIT 2.0 was here constructed as the latest curated dataset focusing on Herbal Ingredients' Targets covering PubMed literatures 2000–2020. Currently, HIT 2.0 hosts 10 031 compound-target activity pairs with quality indicators between 2208 targets and 1237 ingredients from more than 1250 reputable herbs. The molecular targets cover those genes/proteins being directly/indirectly activated/inhibited, protein binders, and enzymes substrates or products. Also included are those genes regulated under the treatment of individual ingredient. Crosslinks were made to databases of TTD, DrugBank, KEGG, PDB, UniProt, Pfam, NCBI, TCM-ID and others. More importantly, HIT enables automatic Target-mining and My-target curation from daily released PubMed literatures. Thus, users can retrieve and download the latest abstracts containing potential targets for interested compounds, even for those not yet covered in HIT. Further, users can log into 'My-target' system, to curate personal target-profiling on line based on retrieved abstracts. HIT can be accessible at <http://hit2.badd-cao.net>.

INTRODUCTION

Being a rich source of drug candidates, herbal active ingredients play a critical role in the development of new drugs. From 1981 to 2019, 33.6% of the drugs approved by the FDA were reported to be derived from natural products or their derivatives (1). To better understand the interaction between herbal compounds and molecular targets, the first herbal-ingredient-target database, HIT, was established in 2010 via manual curation of 1301 literature-described targets for herbal compound from 3250 literatures, with convenient links to therapeutic targets database (TTD) and Drugbank etc (2,3). The target information of HIT has been extensively exploited to study the mechanism of natural compounds, as well as to make discoveries from herbal medicine. For instance, based on HIT, Luo Y et al. revealed the therapeutic mechanism of cryptotanshinone in the treatment of liver cancer (4). And Wang *et al.* identified potential targets for asthma according to the clinical efficacy of TCM formulations (5).

During the past ten years (2011–2021), there was an explosive increase in the studies on the natural ingredients and their targets. A number of databases were then constructed covering herbal compound and target interactions. Notably, a nice exemplary database for natural products, NPASS (6), provides experimentally-determined quantitative activity records for natural products, including nearly 2,000 herbal ingredient-target pairs for about 700 herbal ingredients. Other herbal databases also included important information of herbal ingredient-target pairs (7–11), but most of them downloaded and incorporated early HIT data, appended with predicted targets, such as HERB, TCMID, TCMSP and SymMap (7–10). These databases have greatly enriched the target diversity for herbal ingredients. However, the literature-described targets of herbal ingredients

*通信应当寄给谁。电话: +86 21 65980296; 电子邮件: zwcao@fudan.edu.cn 通信也可寄给唐凯琳。电话:

+86 21 65980296; 电子邮件: kltang@tongji.edu.cn

[†]作者希望声明, 他们认为前两位作者应被视为共同第一作者。

*To whom correspondence should be addressed. Tel: +86 21 65980296; Email: zwcao@fudan.edu.cn

Correspondence may also be addressed to Kailin Tang. Tel: +86 21 65980296; Email: kltang@tongji.edu.cn

[†]The authors wish it to be known that, in their opinion, the first two authors should be regarded as joint First Authors.

虽然经过精心策划，但有价值，但仍有限制，需要持续且持续更新。

然而，这些关于成分-靶点相互作用的新证据可能会发表在日常发布的文献中，而传统数据库很难跟踪最新的研究成果。用户通常需要花费大量精力手动搜索当前数据库尚未涵盖的最新文献，并挖掘靶点以与数据库条目对应，这非常耗时。因此，有必要提出一个新平台，不仅提供定期更新的靶点信息，还能实时检查每日发布的论文中潜在的靶点。

在此，我们为上述目的介绍了HIT 2.0平台。在这个版本中，全面采用了先进的文本挖掘算法和严格的人工整理。除了校准2000年至2010年文献中的HIT 1.0数据外，我们的整理团队还通过添加2010年至2020年的文献进行了全面更新，使数据量几乎增加了一倍，并增加了新的靶点可信度指标功能。更重要的是，化合物-靶点建议的文本挖掘系统现已向用户开放，研究人员可以通过“靶点挖掘”检索任何天然化合物的相关文献。最后，HIT 2.0提供了“我的靶点”的在线功能，方便个人整理和下载。

材料与方法

数据来源

与 HIT 1.0 类似，成分信息来源于广泛使用的 2020 年更新版本 TCM-ID 数据库（12），该数据库涵盖 2751 种药材和 7375 种药材成分。化合物别名来源于化学文摘社（CAS）。用于描述化合物与分子靶点之间相互作用的关键词集合与 HIT 1.0 相同，共 59 个（补充表 S1）。其中一些关键词是描述相互作用的名词（类型 A），而另一些（类型 B）是描述特定作用的短语，如“抑制蛋白质活性”。

文本挖掘

使用自然语言处理（NLP）进行文本挖掘，以识别文献中草药成分的靶点，工作流程如下图所示（图1）：

步骤1：在PubMed中检索包含草药成分名称/别名关键词的摘要。

步骤2：使用PubTator Central (13) 对摘要中的基因/蛋白实体进行注释，该平台是一个用于生物医学实体的自动化注释平台。这些基因/蛋白可能是成分的靶点，且只保留包含基因/蛋白实体的摘要。

步骤3：将完整的摘要解析为句子。根据以下两条规则筛选句子。规则1：‘化合物名称’且‘类型A中的任意词’且‘基因’。例如，句子‘EGCG是一种新型Hsp90抑制剂’。规则2：‘化合物名称’且‘类型B相互作用中的任意词’且‘类型B效应中的任意词’且‘基因’。例如，句子‘原花青素B2直接抑制MT1-MMP活性。’

第4步：使用斯坦福解析器引擎（14）提取句子的句法结构和语法关系（15）。仅保留在定向依赖树路径中检测到“compound”、“gene”和“keyword”的句子。

手动整理

最终，一个由11名博士研究生组成的策展团队审查了17,000个句子，每一项都至少由两名候选人进行复核。两位策展人一致同意的条目被保留。对于存在分歧的条目，由第三名策展人进行审查，最终结果通过多数票决定。

结果

HIT 2.0 的构建基于文献挖掘和人工整理。在文献挖掘部分，首先检索了 PubMed 摘要，包含草药成分名称的关键词或别名。然后将摘要解析为句子，仅保留包含草药成分、基因及相关关键词的句子以供进一步整理。为了方便，每位整理者都有自己的在线平台账户。登录后，整理者可以查看已完成和待完成的 PubMed 任务。目前，整个平台已开放为“Target-mining”和“My-target”系统，使用户能够识别最相关的文献，并密切跟踪其感兴趣化合物的最新靶点。

数据更新

在这个版本中，使用与 HIT 1.0 相同的一套关键词（16），应用了更先进的自然语言处理（NLP）文本挖掘算法和严格的人工整理。NLP 已被广泛应用于生物医学文本挖掘。依赖句法分析可以解析句子结构，从而突显医学实体之间的关系（15）。HIT 2.0 中使用 NLP 算法来确定化合物是否可能与基因/蛋白质相互作用。最初，通过文本挖掘系统从 PubMed 获得了 7100 篇摘要。经过在线整理后，共生成了 10,031 个化合物-靶点对，涉及 2208 个分子靶点和来自 1250 种中药的 1237 种化合物。有趣的是，我们的靶向列表中还包括了 56 个 miRNA 基因，主要以上调/下调基因的方式作用。

化合物-靶点相互作用的类型涵盖了10类：间接抑制/激活，上调/下调基因，直接抑制/激活，结合剂，酶底物，酶产物及其他。已开发用于化合物-靶点相互作用的质量指标，涵盖相互作用的性质、支持同一配对的文献数量以及每篇文献的引用报告。基于上述信息，用户可以选择偏好的相互作用进行进一步分析。每个化合物-靶点配对都可以查看从来源文献解析出的关键描述。化合物已链接到 PubChem（17）和 ChEMBL（18）数据库。至于靶点，已与以下数据库进行了交叉链接

being carefully curated, though valuable, remain limited and need to be updated urgently and regularly.

Yet those fresh evidences of ingredient-target interactions may be published in everyday-released literatures, while the traditional databases have difficulties in keeping track of the latest results. Users often need to take great efforts manually searching the up-to-date literatures not yet covered by current databases and dig targets out to tie up with database items, which is highly time-consuming. Thus, it is necessary to propose a new platform to provide not only the regularly updated targets, but also the real-time checking for potential target from daily-released papers.

Here, we introduced such a platform of HIT 2.0 for the above purpose. In this version, advanced text mining algorithms and rigorous curation were comprehensively employed. In addition to calibrating HIT 1.0 data from literatures between 2000 and 2010, our curation team made a complete refreshment by adding literatures between 2010 and 2020, resulting in almost twice the data abundancy as before, plus new features of target confidence indicators. More importantly, the text-mining system of compound-target suggestion is now open to users, where researchers can retrieve the most-related literatures via ‘Target-mining’ for any natural compounds. At last, HIT 2.0 provides an on-line function of ‘My-target’ for personal curation and downloading.

MATERIALS AND METHODS

Data source

Similar to HIT 1.0, the ingredient information was sourced from the widely used TCM-ID database of the 2020 updated version (12), which covers 2751 herbs and 7375 herb ingredients. Compound aliases were derived from the Chemical Abstracts Service (CAS). The same set of 59 keywords was used as that in HIT 1.0 to describe interactions between compounds and molecular targets (Supplementary Table S1). Some keywords are nouns describing the interaction (Type A), while the others (Type B) are phrases describing the specific effect, such as ‘inhibit the activity of’ proteins.

Text mining

Text mining of NLP was used to identify the targets of herbal ingredients in the literature, with workflow being illustrated as follows (Figure 1):

Step1: Retrieve abstracts in PubMed containing keywords of the name/alias of herb ingredients.

Step2: Annotate gene/protein entities in the abstract using PubTator Central (13), an automated annotation platform for biomedical entities. These genes/proteins may be targets of ingredients, and only abstracts containing gene/protein entities will be retained.

Step3: Parse the full abstract into sentences. Screen out those sentences based on either of below two rules. Rule1: ‘Compound name’ AND ‘any word in type A’ AND ‘Gene’. For instance, the sentence ‘EGCG is a novel Hsp90 inhibitor’. Rule2: ‘Compound name’ AND ‘any word in type B interaction’ AND ‘any word in type B effect’ AND ‘Gene’. For instance, the sentence ‘Procyanidin B2 directly inhibited MT1-MMP activity.’

Step4: Use the Stanford Parser engine (14) to extract sentence syntactic structure and grammatical relations (15). Only those sentences are retained where ‘compound’, ‘gene’ and ‘keyword’ are detected in a directional dependency tree path.

Manual curation

Finally, a curation team of 11 Ph.D. candidates reviewed 17 000 sentences, and each item was double checked by at least two candidates. The entries with consistence from two curators were remained. While those of disagreement were reviewed by a third curator, with the final result being decided by a majority vote.

RESULTS

The construction of HIT 2.0 was based on literature mining and manual curation. In the part of literature mining, PubMed abstracts were firstly retrieved containing keywords of the name/alias of herbal ingredients. Then the abstract was parsed into sentences and only those sentences containing herbal ingredients, genes, and keywords were kept for further curation. For convenience, an online platform was set up where each curator has their own account. Any time after log-in, curators can review the PubMed tasks completed, and to-be-completed. Now the whole platform has been opened as ‘Target-mining’ and ‘My-target’ system, enabling users to identify those most related literatures and keep close tracking of the latest targets for their interested compounds.

Data updating

In this version, more advanced text-mining algorithms of Natural language processing (NLP) and rigorous manual curation were applied using the same set of keywords as that of HIT 1.0 (16). NLP has been widely applied to biomedical text mining. Dependent syntactic analysis enables sentence structure parsing to highlight the relationships between medical entities (15). The NLP algorithm is used in HIT 2.0 to determine whether compounds may interact with genes/proteins. Initially, 7100 abstracts were obtained from PubMed after text-mining system. After the online curation, a total of 10 031 compound-target pairs were produced, involving 2208 molecular targets and 1,237 compounds from 1250 herbs. Interestingly, 56 miRNA genes have been included into our targeting list, mainly with modes of up/down regulated genes.

The types of compound-target interactions cover 10 categories: indirectly inhibit/activate, up/down-regulated gene, directly inhibit/activate, binders, enzyme substrates, enzyme products and other. Quality indicators of compound-target interactions have been developed covering the nature of the interaction, the number of literatures supporting the same pair, and the citing reports of each literature. Based on above, users can choose those preferred for further analysis. Each compound-target pair can be viewed with a key description parsed from the sourcing literature. Compounds are linked to PubChem (17) and ChEMBL (18) databases. As for targets, Crosslinks have been made to databases of

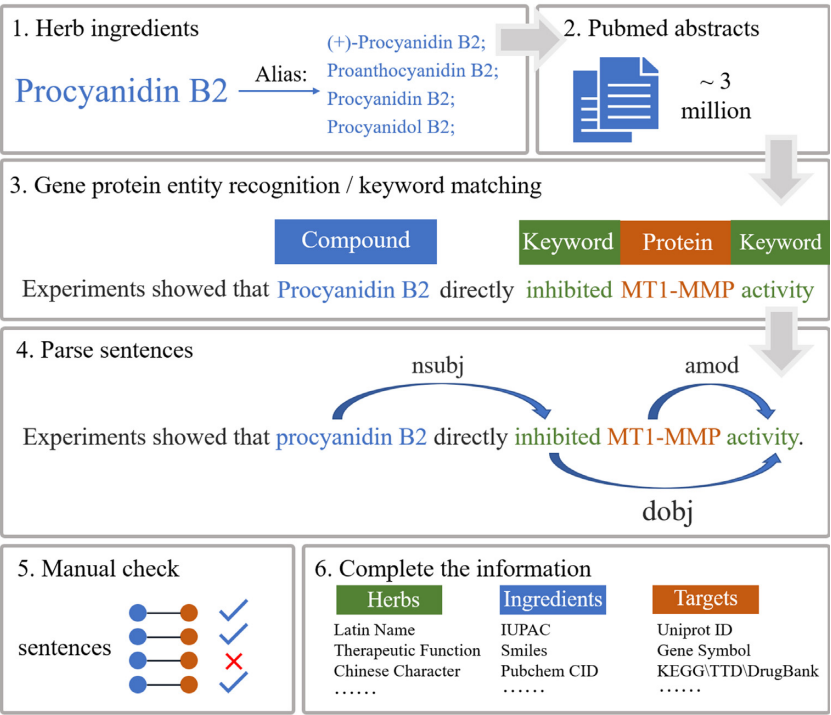


图 1. HIT 2.0 的工作流程。(1, 2) 使用草本成分的不同名称检索 PubMed。(3) 挖掘 PubMed 摘要以识别基因-蛋白实体。(4) 检测“化合物”、“基因”和“关键词”是否在方向依赖树路径中。(5&6) 手动检查并完善信息。

TTD (2)、DrugBank (3)、KEGG (19)、PDB (20)、UniProt (21)、Pfam (22)、NCBI、TCM-ID (12) 等提供更详细的信息。

与包含文献描述靶点的类似草药数据库相比, HIT 2.0 将 HIT 1.0 的数据丰富度提高了一倍, 为之前的数据库提供了很好的补充。目前, 就草药的整理成分-靶点对而言, 它也是最大的数据库, 如表 1 所示。

HIT 2.0 允许关键词搜索和复合相似性搜索。搜索界面和结果页面如图 2 所示。关键词搜索可用于药材信息(中文拼音、汉字、拉丁名和英文名)、成分信息(不同名称、CAS 号和 CID 号)以及靶点信息(基因、蛋白名称、基因符号、UniProt ID)。关键词搜索支持自动完成和模糊搜索。此外, 还可以通过化合物结构进行相似性搜索, 以 Tanimoto 系数 70% 以上为截止。输入可以使用 S MILES 公式, 也可以使用内置软件 Ketcher (<https://lifescience.opensource.epam.com/ketcher/>) 绘制的结构。

目标挖掘与My-target管理

由于 PubMed 文献每天都会发布, 研究人员需要不断查看 2020 年之后与其感兴趣的化合物相关的最新证据。目前, 还没有可用的软件来挖掘包含潜在靶点-成分相互作用的文献。为了方便整理, 我们为用户提供了两种选择。

目标挖掘功能是内置的, 使用户能够检索从 2010 年到每日更新发表的关于任何化合物的相关 PubMed 摘要(图 3A 和 B)。我们意识到, 通过文本挖掘识别的摘要可能包含误报。因此, HIT 2.0 提供了在线 My-target 策展功能, 以进一步检查来源摘要中的详细句子(图 3C)。包含感兴趣实体的句子已用不同颜色标出, 以便可以轻松识别关键项目及其相互作用类型。此外, 还提供了到 PubMed 的链接, 可查看完整摘要。

简而言之, “目标挖掘”旨在基于我们经验得出的规则结合先进的文本挖掘算法, 高效且精准地检索最相关的文摘。通过这一功能, 用户可以检索并下载最新的文献以便本地查看。或者, 用户可以选择“我的目标整理”在线进行整理。每位用户都有自己的账户。登录后, 用户可以随时查看已完成的任務, 并继续完成整个任务。

讨论

作为对其他药物和药物靶点数据库的补充, 包含文献描述靶点的草药数据库已成为天然产物机制研究的主要来源, 通过提供药用草药、活性化合物与分子靶点在不同实验条件下的丰富信息集合。然而, 随着过去十年中新证据文献的不断积累, 以前的版本需要及时更新以满足迫切需求。在

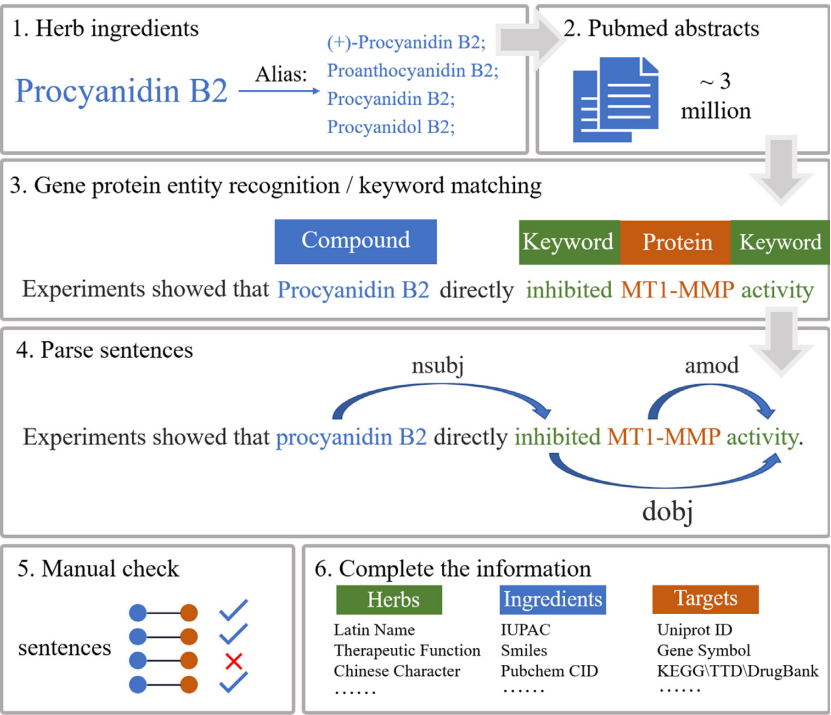


Figure 1. Workflow of HIT 2.0. (1, 2) Retrieve PubMed using different names of herbal ingredients. (3) Mine PubMed abstracts to identify gene/protein entities. (4) Detect whether ‘compound’, ‘gene’ and ‘keyword’ are in a directional dependency tree path. 5&6. Manual check and complete the information.

TTD (2), DrugBank (3), KEGG (19), PDB (20), UniProt (21), Pfam (22), NCBI, TCM-ID (12) and others for more detailed information.

Compared to similar herb databases containing literature-described targets, HIT 2.0 has doubled the data abundance of HIT 1.0, forming a nice complement to the previous databases. Currently, it is also the largest database in terms of curated ingredient-target pairs for herbs, as Table 1 shows.

HIT 2.0 allows keyword search and compound similarity search. The searching interface and results pages are illustrated in Figure 2. Keyword search is available for herb information [Chinese pinyin, Chinese characters, Latin name and English name], ingredients information [different names, CAS number and CID number] and target information [gene/protein name, gene symbol, UniProt ID]. Auto-completion and fuzzy search are supported for keyword search. Besides that, similarity search can also be made via compound structures, with a Tanimoto coefficient above 70% as a cut-off. Both the SMILES formula and the artificially drawn structure by build-in software Ketcher (<https://lifescience.opensource.epam.com/ketcher/>) can be used as input.

Target-mining and My-target curation

As PubMed literatures are released every day, there is a constant need for researchers to check the latest evidences after 2020 for their interested compound. So far, no software becomes available to mine the literatures containing potential target-ingredient interactions. For the convenience of curation, two options have been provided for users.

Target-mining function was built-in enabling users to retrieve related PubMed abstracts published from 2010 till the daily updates for any compounds (Figure 3A and B). It was realized that the abstracts identified by text-mining may contain false positives. Thus HIT 2.0 provides on-line My-target curation function to further check the detailed sentences in sourcing abstracts (Figure 3C). The sentences containing interesting entities have been highlighted in different colors, so that key items can be easily spotted together with the interaction types. A link to PubMed with full abstract is also provided.

In brief, ‘Target-mining’ was designed to efficiently and precisely retrieve those most-related abstracts based on our empirically derived rules combined with advanced text-mining algorithm. Via this, users can retrieve and download the latest literatures for local checking. Alternatively, users can choose ‘My-target curation’ to make curation on-line. Each user has their own account. Anytime after log-in, users can review the tasks completed and continue to finish the full task.

DISCUSSION

Complement to other drug and drug-target databases, herbal databases containing literature-described targets have served as the primary source for mechanistic study of natural products by providing rich sets of information between medicinal herbs, active compounds and molecular targets under different experimental conditions. With a steady literature accumulation describing new evidences in the past decade, however, the previous versions need to make timely updating to meet up with the pressing needs. In

表1. 文献中描述的对等数据库目标概览

	出版年份	文学描述的目标	草本成分	草本成分-靶点活动对	文献搜集
HIT 2.0		2208	1237	10,031	7100篇PubMed摘要
HIT	2011	1301	586	5208	3250 篇 PubMed 摘要
HERB	2020	1241	370	4815	1966 年 PubMed 摘要
TCMID	2013	680	/	/	4500 中文
无通过TCMSP	2017	464	719	1936	1288 篇 PubMed 摘要
	2014	<u>3311 由 SysDT 预测</u>	<u>29,384</u>	<u>84 260 由 SysDT 预测</u>	/

Table 1. Overview of the literature-described targets from peering databases

	Published year	Literature-described targets	Herbal ingredients	Herbal ingredient-target activity pairs	Sourcing literatures
HIT 2.0		2208	1237	10 031	7100 PubMed abstracts
HIT	2011	1301	586	5208	3250 PubMed abstracts
HERB	2020	1241	370	4815	1966 PubMed abstracts
TCMID	2013	680	/	/	4500 Chinese Literatures
NPASS	2017	464	719	1936	1288 PubMed abstracts
TCMSP	2014	<u>3311 predicted by SysDT</u>	<u>29 384</u>	<u>84 260 predicted by SysDT</u>	/

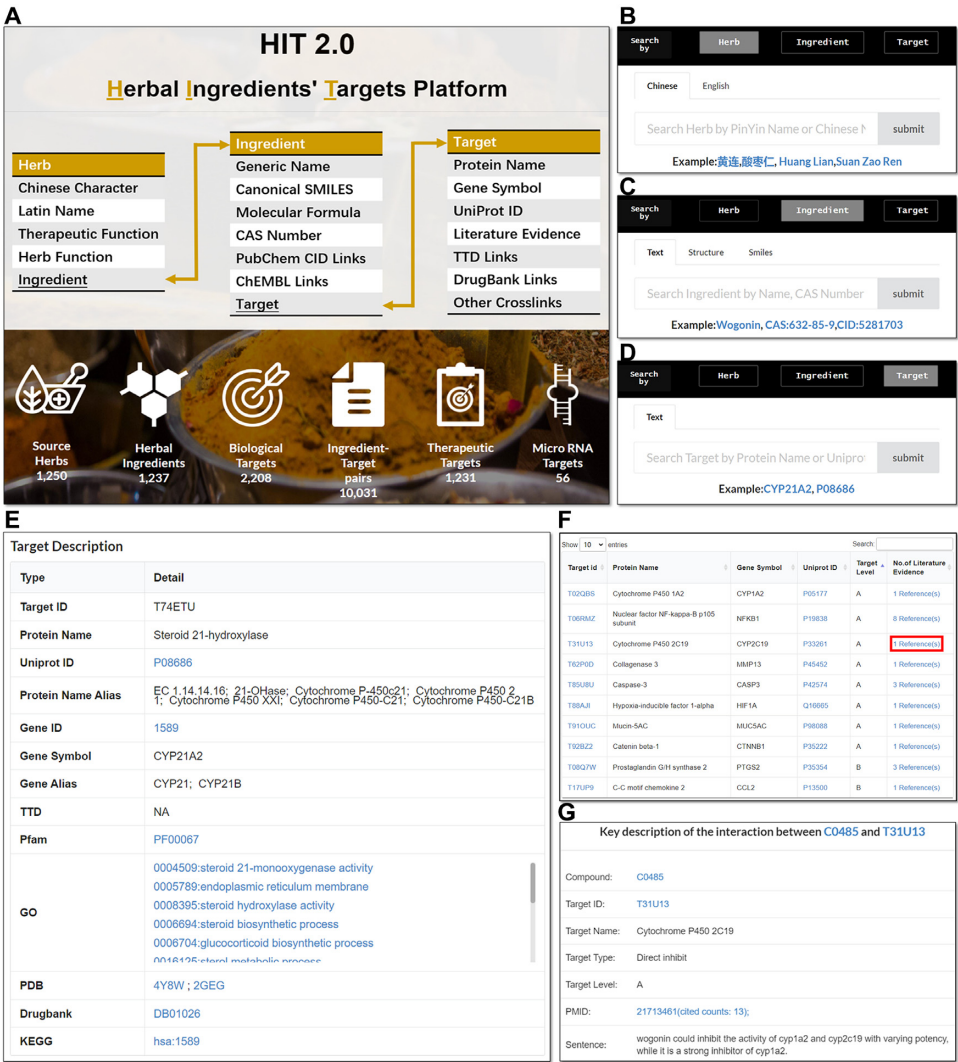


图2. HIT 2.0中的搜索和结果页面。(A) 数据库结构和数据统计。(B) 中药可通过拼音、汉字和拉丁名等关键词进行搜索。(C) 草药成分可通过结构相似性或名称、CID和CAS号关键词进行搜索。(D) 靶点可通过基因/蛋白名称、基因符号和Uniprot ID等关键词进行搜索。(E) 靶点的详细信息。(F) 化合物的其他靶点。(G) “文献证据”提供从相关文献中解析出的关键描述。

因此，本文构建了 HIT 2.0，通过再增加十年的数据来保持定期更新，成为最新且规模最大的关于草药成分-靶点配对的数据库。同时，我们建立了基于自然语言处理技术的靶点挖掘和我的靶点整理系统，使研究人员能够持续追踪最新证据并整理感兴趣的个人靶点。

在方便的时间进行化合物研究。HIT 2.0 的发布将成为通过分子靶点连接草药成分与 FDA 批准药物的重要补充，并可能促进新药可作用分子的发现，以及识别潜在的治疗靶点。

HIT 2.0应用的一个关键技术是自动文本挖掘工具。目前，有几种注释工具

Figure 2. Searching and resulting pages in HIT 2.0. (A) Database structure and data statistics. (B) Herbs can be searched via keywords such as Chinese Pinyin, Chinese characters and Latin names. (C) Herbal ingredients can be searched via structure similarity or keywords of name, CID and CAS number. (D) Targets can be searched via keywords of gene/protein name, gene symbol and Uniprot ID. (E) Detailed information of the targets. (F) Additional targets of the compound. (G) ‘Literature evidence’ provides the key descriptions parsed from sourcing literatures.

this paper, HIT 2.0 was thus constructed to maintain a regular updating by adding another ten years, marking as the latest and the largest one regarding curated ingredient-target pairs for herbs. Meanwhile, we set up Target-mining and My-target curation system based on technology of natural processing language, allowing researchers to keep tracking the latest evidences and curate personal targets of interested

compounds at convenient time. The launch of HIT 2.0 will be an important addition to bridge herbs ingredients and FDA approved drugs via molecular targets and may facilitate the discovery of new druggable molecules, as well as to identify potential therapeutic targets.

One key technology applied in HIT 2.0 was the auto text-mining tools. Currently, there are several annotating

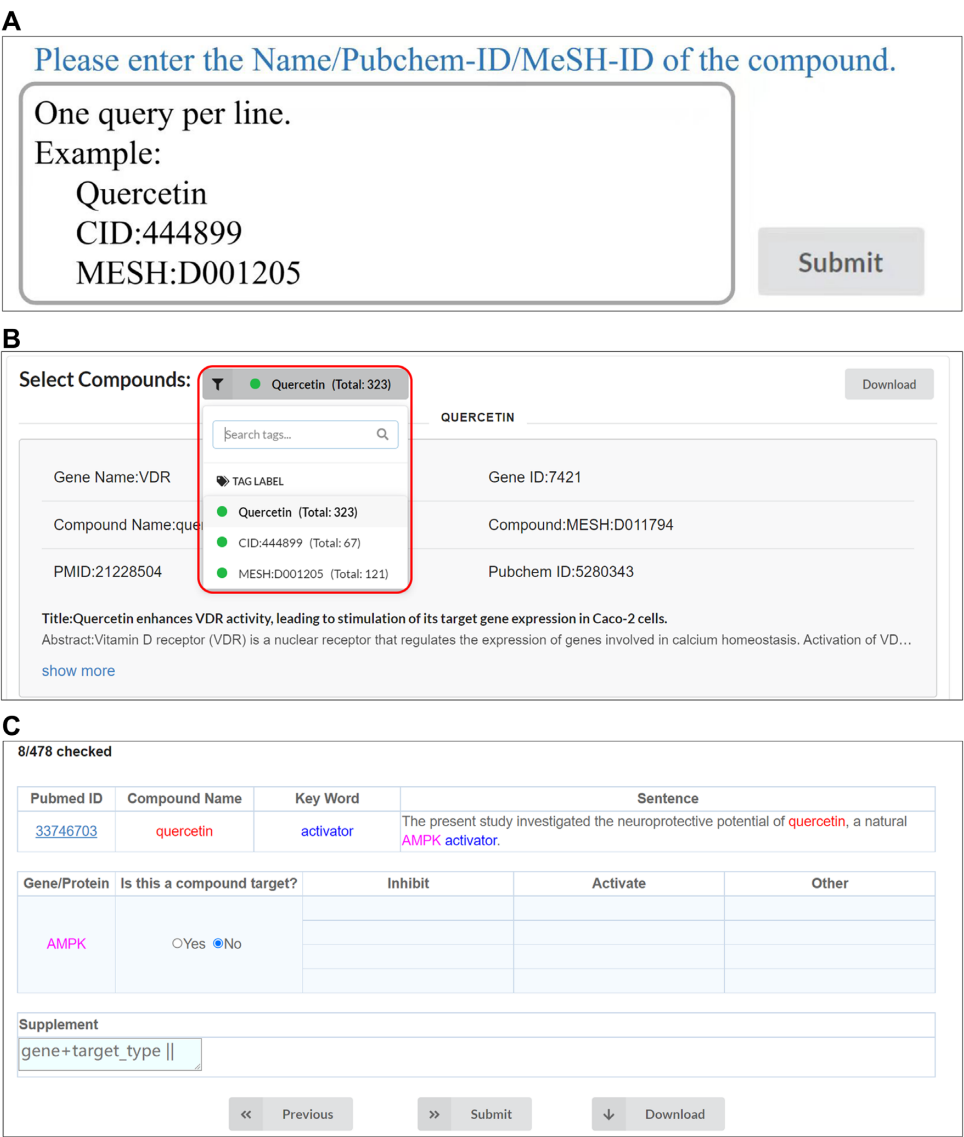


图3. 目标挖掘与我的目标整理系统。(A) 目标挖掘功能界面。可以提交化合物名称、MeSH ID 和 PubChem ID 以检索潜在靶点。(B) 通过目标挖掘检索到的 PubMed 摘要。(C) 我的目标整理系统界面。

用于识别生物医学实体的工具，例如 PubTator Central (13)、HunFlair (23) 和 ScispaCy (24)。其中，PubTator Central 由 NCBI 开发，用于自动标注 PubMed 摘要，并允许用户通过 PubMed ID 列表批量下载标注内容。此外，标注的基因与基因数据库直接链接，这使得 HIT 2.0 可以在无需安装的情况下获取目标的更多详细信息。综合考虑整体便利性、简易性和复杂性，HIT 2.0 选择了 PubTator Central。

与此同时，过去十年中分子靶点在草本化合物功能解释中的应用得到了广泛的发展。尽管文献证据支持的靶点可能比计算预测的靶点更为可靠，但需意识到，收集到 HIT 中的结果大多来自体外实验而非体内实验。事实上，这些天然化合物在体内可能会被代谢成不同形式，从而可能发生变化

靶向谱 (25)。此外，成分的生物活性通常不仅与分子靶点的种类有关，还与靶点之间的网络相互作用有关。从这个意义上说，HIT 2.0 可能是推导草药成分集体功能的有价值的起点，特别是对草药和草药配方。朝着这一方向，HIT 将继续丰富数据的丰富性和后续分析，以保持为领域研究提供高质量资源。

补充数据

补充数据可在 NAR 在线获取。

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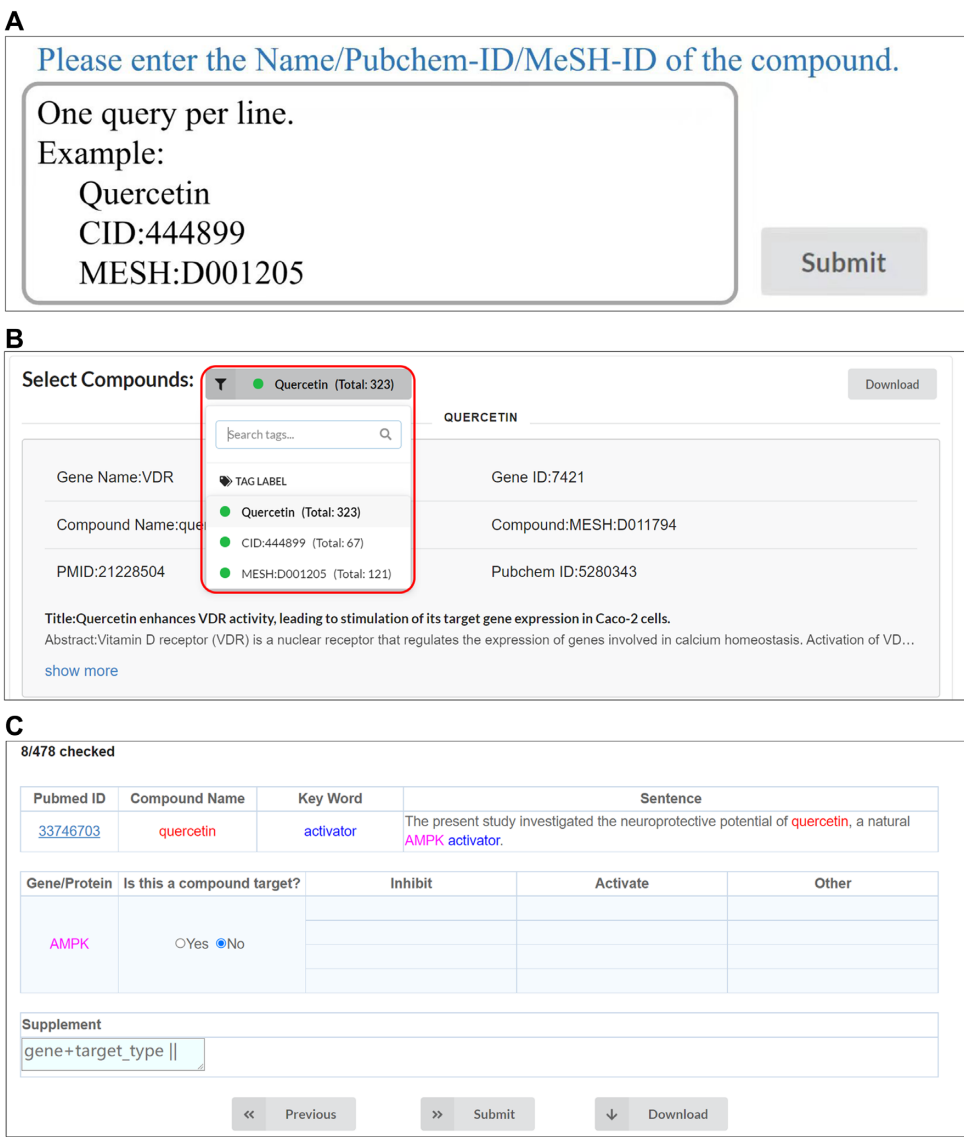


Figure 3. Target-mining and My-target curation system. (A) The interface of Target-mining function. Compound name, MeSH ID and Pubchem ID can be submitted to retrieve potential targets. (B) PubMed abstract retrieved by Target-mining. (C) The interface of My-target curation system.

tools to recognize the biomedical entities, such as PubTator Central (13), HunFlair (23), and ScispaCy (24). Among them, PubTator Central was developed by NCBI to annotate PubMed abstracts automatically and allows users to download annotations in bulk via PubMed ID lists. In addition, the annotated genes have direct links to the Gene database, which allows HIT 2.0 to access more details of the targets without being installed. Considering the overall convenience, simplicity and complexity, PubTator central was chosen for HIT 2.0.

Meanwhile, the last decade has witnessed an extensive application of molecular targets into functional explanation for herbal compounds. Though the reliability of literature-evidenced targets may be more accurate than computationally-predicted targets, it is aware that the results collected into HIT are mostly from *in vitro* rather than *in vivo* experiments. In fact, these natural compounds are likely metabolized into different forms which may change

the targeting profiles (25). Furthermore, the biological activity of an ingredient is often related to not only the group of molecular targets, but also the network interactions among targets. In this sense, HIT 2.0 may serve as a valuable start in deriving collective functions of herbal components, particularly to herbs and herbal formulas. Towards this direction, HIT will continue to enrich the data abundance and subsequent analysis to maintain high-quality resources for domain research.

SUPPLEMENTARY DATA

Supplementary Data are available at NAR Online.

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参考文献

1. Newman, D.J. 和 Cragg, G.M. (2020) 近四十年间从 01/1981 到 09/2019的天然产物作为新药来源。《国家杂志》, 83,770–803。2. 王耀、张志、李富、周、张、张、王梓、张振、朱志、任义、谭等 (2020) 治疗靶点数据库2020：促进靶向治疗研究和早期开发的丰富资源。核酸研究, 48, D1031–D1041。3. Wishart, D.S., Feunang, Y.D., Guo, A.C., Lo, E.J., Marcu, A., Grant, J.R., Sajed, T., Johnson, D., Li, C., Sayeeda, Z. 等 (2018) DrugBank 5.0: 2018年DrugBank数据库的重大更新。核酸研究, 46, D1074–D1082。4. 罗荣、宋、L.、王晓、黄耀、刘耀、王洁、洪、M. 和袁志 (2020) 揭示隐铈醌作为治疗肝细胞癌的机制。前面。《药学》, 第11卷, 1264页。5. Wang, Y., Chen, Y.J., Xiang, C., 江, G.W., 徐, Y.D., Yin, L.M., 周, D.D., Liu, Y.Y. 和 Yang, Y.Q. (2020) 潜在哮喘指标的发现

医学及其在机制研究和处方验证中的应用。《布鲁斯·J·法玛科尔》, 149,1092–1103。13. Wei, C.H., Allot, A., Leaman, R. 和 Lu, Z. (2019) PubTator 中心：生物医学全文文章的自动概念注释。核酸研究, 47, W587–W593。14. Marneffe, M.-C., MacCartney, B. 和 Manning, C. (2006) 载于：从短语结构解析生成类型依赖解析。http://www.lrec-conf.org/proceedings/lrec2006/pdf/440pdf.pdf。15. Kveler, K., Starosvetsky, E., Ziv-Kenet, A., Kalugny, Y., Gorelik, Y., Shalev-Malul, G., Aizenbud-Reshef, N., Dubovik, T., Briller, M., Campbell, J. 等 (2018) 通过PubMed计算挖掘识别出疾病背景中以免疫为中心的细胞因子和细胞网络。《自然生物技术》, 36,651–659。16. 叶浩、叶、姜浩、张大、陶、唐、刘晓、朱、刘启、刘洁等 (2011) HIT：将草本活性成分与靶点联系起来。核酸研究, 39, D1055–D1059。17. 金S., 陈J., 郑T., 金德

access charge: National Key R&D Program of China [2017YFC1700200]。
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REFERENCES

1. Newman,D.J. and Cragg,G.M. (2020) Natural products as sources of new drugs over the nearly four decades from 01/1981 to 09/2019. *J. Nat. Prod.*, **83**, 770–803。2. Wang,Y., Zhang,S., Li,F., Zhou,Y., Zhang,Y., Wang,Z., Zhang,R., Zhu,J., Ren,Y., Tan,Y. *et al.* (2020) Therapeutic target database 2020: enriched resource for facilitating research and early development of targeted therapeutics. *Nucleic Acids Res.*, **48**, D1031–D1041。3. Wishart,D.S., Feunang,Y.D., Guo,A.C., Lo,E.J., Marcu,A., Grant,J.R., Sajed,T., Johnson,D., Li,C., Sayeeda,Z. *et al.* (2018) DrugBank 5.0: a major update to the DrugBank database for 2018. *Nucleic Acids Res.*, **46**, D1074–D1082。4. Luo,Y., Song,L., Wang,X., Huang,Y., Liu,Y., Wang,Q., Hong,M. and Yuan,Z. (2020) Uncovering the mechanisms of cryptotanshinone as a therapeutic agent against hepatocellular carcinoma. *Front. Pharmacol.*, **11**, 1264。5. Wang,Y., Chen,Y.J., Xiang,C., Jiang,G.W., Xu,Y.D., Yin,L.M., Zhou,D.D., Liu,Y.Y. and Yang,Y.Q. (2020) Discovery of potential asthma targets based on the clinical efficacy of Traditional Chinese Medicine formulas. *J. Ethnopharmacol.*, **252**, 112635。6. Zeng,X., Zhang,P., He,W., Qin,C., Chen,S., Tao,L., Wang,Y., Tan,Y., Gao,D., Wang,B. *et al.* (2018) NPASS: natural product activity and species source database for natural product research, discovery and tool development. *Nucleic Acids Res.*, **46**, D1217–D1222。7. Fang,S., Dong,L., Liu,L., Guo,J., Zhao,L., Zhang,J., Bu,D., Liu,X., Huo,P., Cao,W. *et al.* (2021) HERB: a high-throughput experiment- and reference-guided database of traditional Chinese medicine. *Nucleic Acids Res.*, **49**, D1197–D1206。8. Huang,L., Xie,D., Yu,Y., Liu,H., Shi,Y., Shi,T. and Wen,C. (2018) TCMID 2.0: a comprehensive resource for TCM. *Nucleic Acids Res.*, **46**, D1117–D1120。9. Ru,J., Li,P., Wang,J., Zhou,W., Li,B., Huang,C., Li,P., Guo,Z., Tao,W., Yang,Y. *et al.* (2014) TCMSP: a database of systems pharmacology for drug discovery from herbal medicines. *J. Cheminform.*, **6**, 13。10. Wu,Y., Zhang,F., Yang,K., Fang,S., Bu,D., Li,H., Sun,L., Hu,H., Gao,K., Wang,W. *et al.* (2019) SymMap: an integrative database of traditional Chinese medicine enhanced by symptom mapping. *Nucleic Acids Res.*, **47**, D1110–D1117。11. Zhang,R.Z., Yu,S.J., Bai,H. and Ning,K. (2017) TCM-Mesh: The database and analytical system for network pharmacology analysis for TCM preparations. *Sci. Rep.*, **7**, 2821。12. Chen,X., Zhou,H., Liu,Y.B., Wang,J.F., Li,H., Ung,C.Y., Han,L.Y., Cao,Z.W. and Chen,Y.Z. (2006) Database of traditional Chinese

medicine and its application to studies of mechanism and to prescription validation. *Br. J. Pharmacol.*, **149**, 1092–1103。13. Wei,C.H., Allot,A., Leaman,R. and Lu,Z. (2019) PubTator central: automated concept annotation for biomedical full text articles. *Nucleic Acids Res.*, **47**, W587–W593。14. Marneffe,M.-C., MacCartney,B. and Manning,C. (2006) In: *Generating Typed Dependency Parses from Phrase Structure Parses*. <http://www.lrec-conf.org/proceedings/lrec2006/pdf/440.pdf.pdf>。15. Kveler,K., Starosvetsky,E., Ziv-Kenet,A., Kalugny,Y., Gorelik,Y., Shalev-Malul,G., Aizenbud-Reshef,N., Dubovik,T., Briller,M., Campbell,J. *et al.* (2018) Immune-centric network of cytokines and cells in disease context identified by computational mining of PubMed. *Nat. Biotechnol.*, **36**, 651–659。16. Ye,H., Ye,L., Kang,H., Zhang,D., Tao,L., Tang,K., Liu,X., Zhu,R., Liu,Q., Chen,Y.Z. *et al.* (2011) HIT: linking herbal active ingredients to targets. *Nucleic Acids Res.*, **39**, D1055–D1059。17. Kim,S., Chen,J., Cheng,T., Gindulyte,A., He,J., He,S., Li,Q., Shoemaker,B.A., Thiessen,P.A., Yu,B. *et al.* (2021) PubChem in 2021: new data content and improved web interfaces. *Nucleic Acids Res.*, **49**, D1388–D1395。18. Mendez,D., Gaulton,A., Bento,A.P., Chambers,J., De Veij,M., Felix,E., Magarinos,M.P., Mosquera,J.F., Mutowo,P., Nowotka,M. *et al.* (2019) ChEMBL: towards direct deposition of bioassay data. *Nucleic Acids Res.*, **47**, D930–D940。19. Kanehisa,M., Furumichi,M., Sato,Y., Ishiguro-Watanabe,M. and Tanabe,M. (2021) KEGG: integrating viruses and cellular organisms. *Nucleic Acids Res.*, **49**, D545–D551。20. Burley,S.K., Bhikadiya,C., Bi,C., Bittrich,S., Chen,L., Crichlow,G.V., Christie,C.H., Dalenberg,K., Di Costanzo,L., Duarte,J.M. *et al.* (2021) RCSB Protein Data Bank: powerful new tools for exploring 3D structures of biological macromolecules for basic and applied research and education in fundamental biology, biomedicine, biotechnology, bioengineering and energy sciences. *Nucleic Acids Res.*, **49**, D437–D451。21. UniProt,C. (2021) UniProt: the universal protein knowledgebase in 2021. *Nucleic Acids Res.*, **49**, D480–D489。22. Mistry,J., Chuguransky,S., Williams,L., Qureshi,M., Salazar,G.A., Sonnhammer,E.L.L., Tosatto,S.C.E., Paladin,L., Raj,S., Richardson,L.J. *et al.* (2021) Pfam: The protein families database in 2021. *Nucleic Acids Res.*, **49**, D412–D419。23. Weber,L., Sanger,M., Munchmeyer,J., Habibi,M., Leser,U. and Akbik,A. (2021) HunFlair: an easy-to-use tool for state-of-the-art biomedical named entity recognition. *Bioinformatics*,**37**, 2792–2794。24. Neumann,M., King,D., Beltagy,I. and Ammar,W. (2019) In: *Proceedings of the 18th BioNLP Workshop and Shared Task*. <https://aclanthology.org/W19-50>。25. Kang,H., Tang,K., Liu,Q., Sun,Y., Huang,Q., Zhu,R., Gao,J., Zhang,D., Huang,C. and Cao,Z. (2013) HIM-herbal ingredients in-vivo metabolism database. *J. Cheminform.*, **5**, 28。